Find LU-decomposition of the coefficient matrix and hence solve the following system of equations:

$$2x + 6y + 2z = 2; \qquad -3x - 8y = 2; \qquad 4x + 9y + 2z = 3.$$

$\begin{bmatrix} 2 & 6 & 2 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	
$R_1' \leftarrow R_1 / 2$	$E_1 = \begin{bmatrix} 1/2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$E_1^{-1} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
$= \begin{bmatrix} 1 & 3 & 1 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{bmatrix}$ $R_2' \leftarrow R_2 + 3R_1$	$E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$E_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
$= \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 4 & 9 & 2 \end{bmatrix}$ $R_3' \leftarrow R_3 - 4R_1$	$E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}$	$E_3^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix}$
$= \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & -3 & -2 \end{bmatrix}$ $R_3' \leftarrow R_3' + 3R_2'$	$E_4 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}$	$E_4^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix}$
$= \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 7 \end{bmatrix}$ $R_3' \leftarrow R_3' / 7$	$E_5 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1/7 \end{bmatrix}$	$E_5^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 7 \end{bmatrix}$
$= \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$		

$L = E_1^{-1} \quad E_2^{-1} \quad E_3^{-1} \quad E_4^{-1} \quad E_5^{-1}$

$$= \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 7 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -3 & 7 \end{bmatrix}$$

Therefore, $\begin{bmatrix} 2 & 6 & 2 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -3 & 7 \end{bmatrix} \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$

So, $AX = B$ can be written as

$$\begin{bmatrix} 2 & 6 & 2 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -3 & 7 \end{bmatrix} \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix} \text{.........................} (1)$$

We know, $UX = Y$

$$\begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} p \\ q \\ r \end{bmatrix} \text{..} (2)$$

Therefore (1) becomes

$$\begin{bmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -3 & 7 \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$$

$$\Rightarrow \begin{aligned} 2p \qquad\qquad &= 2 \\ -3p+q \qquad &= 2 \\ 4p-3q+7r &= 3 \end{aligned}$$

Forward Substitution

$$\therefore \quad p = 1; \quad q = 5; \; r = 2$$

From (2)

$$\begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix}$$

$$\Rightarrow \begin{aligned} x+3y+\; z &= 1 \\ y+3z &= 5 \\ z &= 2 \end{aligned}$$

$$\therefore \quad z = 2; \quad y = -1; \; x = 2$$

◆ Example 1 (2×2 System)

Solve the system

$$x + y = 3, \quad 2x + 3y = 7$$

using LU decomposition.

Step 1: Write the system in matrix form

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} 3 \\ 7 \end{bmatrix}.$$

So, $AX = B$.

Step 2: LU decomposition (Doolittle's method, L has 1s on diagonal)

We want $A = LU$, where

$$L = \begin{bmatrix} 1 & 0 \\ l_{21} & 1 \end{bmatrix}, \quad U = \begin{bmatrix} u_{11} & u_{12} \\ 0 & u_{22} \end{bmatrix}.$$

Comparing:

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ l_{21} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} \\ 0 & u_{22} \end{bmatrix}.$$

Multiply:

$$= \begin{bmatrix} u_{11} & u_{12} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} \end{bmatrix}.$$

Match entries:

- $u_{11} = 1$
- $u_{12} = 1$
- $l_{21}u_{11} = 2 \Rightarrow l_{21} = 2$
- $l_{21}u_{12} + u_{22} = 3 \Rightarrow 2 \cdot 1 + u_{22} = 3 \Rightarrow u_{22} = 1.$

So,

$$L = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

◆ Example 2 (3×3 System)

Solve

$$x + y + z = 6, \quad 2x + 3y + z = 10, \quad x + 2y + 3z = 13.$$

Step 1: Matrix form

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 1 & 2 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 6 \\ 10 \\ 13 \end{bmatrix}.$$

Step 2: LU decomposition (Crout's method, U has 1s on diagonal)

Assume

$$L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}, \quad U = \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}.$$

Multiply L and U:

$$LU = \begin{pmatrix} \ell_{11} & \ell_{11}u_{12} & \ell_{11}u_{13} \\ \ell_{21} & \ell_{21}u_{12} + \ell_{22} & \ell_{21}u_{13} + \ell_{22}u_{23} \\ \ell_{31} & \ell_{31}u_{12} + \ell_{32} & \ell_{31}u_{13} + \ell_{32}u_{23} + \ell_{33} \end{pmatrix}.$$

This must equal A. Equate entries row-wise to get equations for the unknowns.

Step 1 — first row gives ℓ_{11}, u_{12}, u_{13}

From row 1:

$$\ell_{11} = 1, \qquad \ell_{11}u_{12} = 1, \qquad \ell_{11}u_{13} = 1.$$

Since $\ell_{11} = 1$, we get

$$u_{12} = 1, \qquad u_{13} = 1.$$

Step 2 — second row gives $\ell_{21}, \ell_{22}, u_{23}$

From row 2 we must match $[2\ 3\ 1]$. So

$$\ell_{21} = 2,$$

$$\ell_{21}u_{12} + \ell_{22} = 3,$$

$$\ell_{21}u_{13} + \ell_{22}u_{23} = 1.$$

Substitute knowns $\ell_{21} = 2,\ u_{12} = 1,\ u_{13} = 1$:

$$2 \cdot 1 + \ell_{22} = 3 \ \Rightarrow\ \ell_{22} = 1.$$

Next,

$$2 \cdot 1 + 1 \cdot u_{23} = 1 \ \Rightarrow\ 2 + u_{23} = 1 \ \Rightarrow\ u_{23} = -1.$$

Step 3 — third row gives $\ell_{31}, \ell_{32}, \ell_{33}$

From row 3 we must match $[1\ 2\ 3]$. So

$$\ell_{31} = 1,$$

$$\ell_{31}u_{12} + \ell_{32} = 2,$$

$$\ell_{31}u_{13} + \ell_{32}u_{23} + \ell_{33} = 3.$$

Substitute knowns $\ell_{31} = 1,\ u_{12} = 1,\ u_{13} = 1,\ u_{23} = -1$:

$$1 \cdot 1 + \ell_{32} = 2 \ \Rightarrow\ \ell_{32} = 1,$$

$$1 \cdot 1 + 1 \cdot (-1) + \ell_{33} = 3 \ \Rightarrow\ 1 - 1 + \ell_{33} = 3 \ \Rightarrow\ \ell_{33} = 3.$$

So the factors are

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 1 & 3 \end{pmatrix}, \qquad U = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Exercise

1. Use the method of Example 1 and the LU-decomposition

$$\begin{bmatrix} 3 & -6 \\ -2 & 5 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$$

to solve the system

$$\begin{aligned} 3x_1 - 6x_2 &= 0 \\ -2x_1 + 5x_2 &= 1 \end{aligned}$$

2. Use the method of Example 1 and the LU-decomposition

$$\begin{bmatrix} 3 & -6 & -3 \\ 2 & 0 & 6 \\ -4 & 7 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 0 \\ 2 & 4 & 0 \\ -4 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

to solve the system

$$\begin{aligned} 3x_1 - 6x_2 - 3x_3 &= -3 \\ 2x_1 \qquad + 6x_3 &= -22 \\ -4x_1 + 7x_2 + 4x_3 &= 3 \end{aligned}$$

► In Exercises 3–6, find an LU-decomposition of the coefficient matrix, and then use the method of Example 1 to solve the system. ◄

3. $\begin{bmatrix} 2 & 8 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$

4. $\begin{bmatrix} -5 & -10 \\ 6 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -10 \\ 19 \end{bmatrix}$

5. $\begin{bmatrix} 2 & -2 & -2 \\ 0 & -2 & 2 \\ -1 & 5 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -4 \\ -2 \\ 6 \end{bmatrix}$

6. $\begin{bmatrix} -3 & 12 & -6 \\ 1 & -2 & 2 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -33 \\ 7 \\ -1 \end{bmatrix}$

► In Exercises 7–8, an LU-decomposition of a matrix A is given.

(a) Compute L^{-1} and U^{-1}.

(b) Use the result in part (a) to find the inverse of A. ◄

7. $A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 2 & 1 \\ -6 & -1 & 2 \end{bmatrix}$;

$$A = LU = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -3 & -1 & 1 \end{bmatrix} \begin{bmatrix} 2 & -1 & 3 \\ 0 & 4 & -5 \\ 0 & 0 & 6 \end{bmatrix}$$

8. The LU-decomposition obtained in Example 2.